

Design Calculation Sheet — V10a_final_h45

Inputs

Item	Value	Unit	Formula / source
Positive electrode thickness	75.6	um	Chen2020 set (dumped effective value)
Negative electrode thickness	85.2	um	Chen2020 set
Separator thickness	10.0	um	params_r6_v10a_h45.json (base 12)
Positive current collector thickness	8.0	um	params_r6_v10a_h45.json (base 16)
Negative current collector thickness	6.0	um	params_r6_v10a_h45.json (base 12)
Positive electrode porosity	0.4	—	params_r6_v10a_h45.json
Negative electrode porosity	0.35	—	params_r6_v10a_h45.json
Separator porosity	0.47	—	Chen2020 set
Positive electrode density	3262.0	kg/m3	Chen2020 set
Negative electrode density	1657.0	kg/m3	Chen2020 set
Separator density	397.0	kg/m3	Chen2020 set
Electrode height / width	0.065 / 1.58	m	Chen2020 set
Electrode area	0.10270000000000001	m2	height x width
Nominal cell capacity	5.0	Ah	Chen2020 set
Voltage window	2.5 - 4.2	V	Chen2020 set cut-offs
Electrolyte conductivity (design)	4.0	S/m	params_r6_v10a_h45.json (override)
Electrolyte diffusivity (design)	3.539e-10	m2/s	params_r6_v10a_h45.json (override)
Cation transference number (design)	0.45	—	params_r6_v10a_h45.json (override)
Total heat transfer coefficient (design)	45.0	W/m2K	params_r6_v10a_h45.json (override)

Capacity_Energy

Item	Value	Unit	Source
1C discharge capacity	5.06181405203375	Ah	r6_final_energy_dfn.json:capacity_ah
Discharge energy (trapezoid V·I_1C dt)	18.400033998292137	Wh	r6_final_energy_dfn.json:energy_wh
Discharge midpoint voltage	3.8877845799528186	V	r6_final_energy_dfn.json:midpoint_voltage_v
DC resistance	3.3670380507965447	mOhm	r6_final_energy_dfn.json:dcr_ohm
Power density (V_OC^2/4R / mass)	38491.35321522924	W/kg	r6_final_energy_dfn.json:power_density_w_kg

Energy_Density

Item	Value	Unit	Formula / source
Cell mass (contract caliber, electrolyte excluded)	32.575729316	g	r6_final_energy_dfn.json:mass_kg
Gravimetric energy density	564.8387429734295	Wh/kg	energy_wh / mass_kg
Stack thickness	184.8	um	sum of 5 layer thicknesses
Stack volume	18.978960000000004	cm3	thickness x area
Volumetric energy density	969.4964317482167	Wh/L	energy_wh / volume (electrolyte/casing excluded)
Criterion ED >= 327.18 Wh/kg	PASS	—	564.84 >= 327.18
Criterion ED_vol >= 880 Wh/L	PASS	—	969.50 >= 880

Mass_Balance

Layer	Mass (g)	Source
Positive electrode solids	15.195935664	layer_kg_m2.positive_electrode x area
Negative electrode solids	9.424230582	layer_kg_m2.negative_electrode x area
Positive current collector (Al)	2.2183200000000003	layer_kg_m2.positive_cc x area
Negative current collector (Cu)	5.5211520000000001	layer_kg_m2.negative_cc x area
Separator	0.21609107000000005	layer_kg_m2.separator x area
Total (contract)	32.575729316	r6_final_energy_dfn.json:mass_kg

Layer	Mass (g)	Source
Electrolyte (estimate)	7.9810224000000005	pore volume 6.651 cm3 x 1.2 g/cm3 (literature density)
N/P ratio	Not provided	Chen2020 has no per-electrode capacity-density keys; balance unchanged from baseline

Process

Item	Value	Unit	Formula
Positive areal density	147.96	g/m2	75.6e-6 x (1-0.40) x 3262
Negative areal density	91.76	g/m2	85.2e-6 x (1-0.35) x 1657
Positive compaction density	1.957	g/cm3	3262 x 0.60 / 1000
Negative compaction density	1.077	g/cm3	1657 x 0.65 / 1000
Electrolyte fill amount	7.9810224000000005	g	pore volume x 1.2 g/cm3 x 1.0 fill factor
Formation recommendation	0.1C CC to 4.2 V, 25 C, 2 cycles	—	design recommended value; production-line value requires tuning